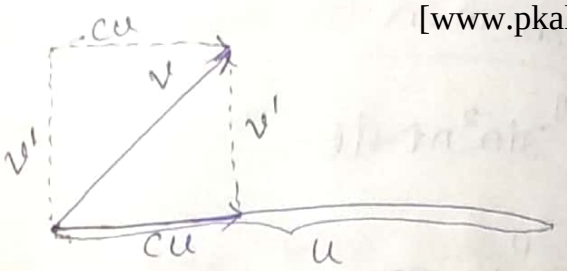
$v = v' + cu$

$v' = v - cu$



$\langle v', u \rangle = \langle v - cu, u \rangle = 0$

$\Rightarrow \langle v, u \rangle - c\langle u, u \rangle = 0$

$\Rightarrow \boxed{c = \frac{\langle v, u \rangle}{\langle u, u \rangle}}$ {Projection of v on $u = cu$}

* Let $u \in \mathbb{R}^2$ st $u = (a, b)$

$\langle v, u \rangle = 0$

$(x, y) \cdot (a, b) = 0$

$\Rightarrow ax + by = 0$

Orthogonal Compliment :-

Let S be a subset of an innerproduct space V. The orthogonal compliment of S is denoted by $S^\perp$ and defined as $S^\perp = \{ v \in V : \langle v, u \rangle = 0 \text{ for every } u \in S \}$

Orthogonal compliment of a single vector

$u^\perp = \{ v \in V : \langle v, u \rangle = 0 \}$

Pkalika Maths